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Formation of Protoplanets from Planetesimals in the Solar Nebula

Eiichiro Kokubo

Department of Earth Science and Astronomy, University of Tokyo, Komaba, Meguro-ku, Tokyo, 153-8902, Japan
E-mail: kokubo@chianti.c.u-tokyo.ac.jp

and

Shigeru Ida

Department of Earth and Planetary Sciences, Tokyo Institute of Technology, Ookayama, Meguro-ku, Tokyo, 152-8551, Japan

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Planetary accretion from planetesimals to protoplanets is investigated using three-dimensional N-body simulations. The effect of gas drag due to solar nebula is included and realistic-sized planetesimals with a standard material density are used, with which the growth time scale of planetesimals is realistic. In agreement with the earlier works (E. Kokubo and S. Ida 1996, *Icarus* 123, 180–191; 1998, *Icarus* 131, 171–178) that investigated gas-free accretion of planetesimals with enlarged sizes, it is found that a bimodal protoplanet–planetesimal system is formed through runaway and oligarchic growth. In the intermediate accretion stage, the growth mode of planetesimals is runaway growth where the mass distribution relaxes into isolated runaway bodies and the continuous power-law mass distribution with $dn_c/dm \propto m^\alpha$, where n_c is the cumulative number of bodies and $\alpha \simeq -2.5$. While thinning out some runaway bodies, the growth mode shifts to oligarchic growth where larger protoplanets tend to grow more slowly than smaller ones. The orbital separations of the protoplanets are kept wider than about 5 Hill radii of the protoplanets through orbital repulsion. In the late accretion stage, similar-sized protoplanets grow, while most planetesimals remain small. Protoplanets with mass 10^{26} g are formed at 1 AU in 500,000 years. © 2000 Academic Press

Key Words: **planetary formation; planetary accretion; planetesimals; N-body simulation.**

1. INTRODUCTION

In the standard scenario of planetary formation, terrestrial planets and the cores of jovian planets are formed through accretion of many small bodies called planetesimals whose initial masses are 10^{18-21} g (e.g., Safronov 1969, Hayashi *et al.* 1985). This process is called planetary accretion.

Planetary accretion has been investigated mainly by solving the coagulation equation of planetesimals under the particle-in-a-box approximation (e.g., Greenberg *et al.* 1978, Nakagawa *et al.* 1983, Wetherill and Stewart 1989, Barge and Pellat 1991, Weidenschilling *et al.* 1997). Evolution of the mass distribution of planetesimals is calculated by using the collision probability given as a function of mass and velocity of planetesimals, coupling with the calculation of the evolution of the velocity distribution by using the kinetic theory. By this approach, it is found that in the intermediate stage of planetary accretion, a larger planetesimal grows faster than smaller ones, resulting in the runaway growth of the largest planetesimal (Greenberg *et al.* 1978, Wetherill and Stewart 1989). This is because in the planetesimal system, the larger a planetesimal becomes, the larger its growth rate becomes by gravitational focusing and dynamical friction from smaller planetesimals. The merit of the statistical approach is to be able to handle a huge number of bodies. The coagulation equation, however, does not include the position of bodies; in other words, the uniform spatial distribution is assumed. This assumption holds in the early stages of planetary accretion. However, it breaks down once the spatial structure is formed by predominantly growing runaway bodies. Therefore, it is generally difficult to investigate the late stage of planetary accretion using the coagulation equation. There are a few calculations that introduced the spatial distribution in the coagulation equation. For example, Weidenschilling *et al.* (1997) used multizoned coagulation equations.

The complementary method to the statistical approach is N-body simulation. In N-body simulation, the gravitational interaction of bodies are exactly evaluated and then the orbits of bodies are numerically integrated. As the evolution of the orbits of individual bodies are calculated, this approach is suitable for the study of the late stage of planetary accretion. However, the high calculation cost proportional to the square of the number of bodies limits the number of available bodies.

By using the special-purpose computer for N-body simulation, GRAPE (Sugimoto *et al.* 1990), which is the pipeline for the fast calculation of gravitational interaction of bodies, Ida and Makino (1992a,b) investigated the evolution of the velocity distribution of planetesimals and found that dynamical

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friction effectively damps the velocity of large planetesimals in their accretion time scale. Ida and Makino (1993) found that the velocity dispersion of planetesimals in the vicinity of a protoplanet is pumped up mainly by the protoplanet when its mass ratio to the mass of palentesimals is larger than a critical value, which leads to slowdown of the runaway growth of the protoplanet.

We have also performed a series of N-body simulation by using the special-purpose computer HARP/GRAPE (Makino *et al.* 1997). Kokubo and Ida (1996) confirmed runaway growth and investigated the evolution of a planetesimal system in detail. The orbital evolution of protoplanets in a swarm of planetesimals was studied by Kokubo and Ida (1995), where they found the orbital repulsion of protoplanets. Then, the post-runaway stage was investigated by Kokubo and Ida (1998). They found oligarchic growth of protoplanets, which means that similar-sized protoplanets grow keeping their orbital separations proportional to their Hill radii through the orbital repulsion.

All these N-body simulations treated gas-free cases. This is because to see the basic process of planetesimal dynamics as the first step, simple models are better than the complex model and the time span of the simulations is shorter than the time scale of gas drag. However, it is necessary to include the effect of the solar nebula gas in the study of the long-term evolution of planetesimal systems. Many young stars have been found to have the gas disk, and, at least, jovian planets must be formed in the solar nebula since they have massive gas envelopes. The effect of gas drag on planetary accretion was investigated by the statistical method (e.g., Nakagawa *et al.* 1983, Ohtsuki *et al.* 1988, Wetherill and Stewart 1989, Barge and Pellat 1991, Spaute *et al.* 1991), which found that the growth mode of planetesimals is only weakly affected by gas drag. In the present paper, we extend N-body simulation to include the drag force on planetesimals by the solar nebula gas.

We also adopt the realistic volume density of planetesimals, in other words, no radius enhancement, which was used in Kokubo and Ida (1996, 1998). Although the previous papers showed that the artificial radius enhancement does not change the growth mode of planetesimals, its direct proof is necessary. Furthermore, with no radius enhancement, the growth time scale obtained by the simulation is realistic.

The effect of the gas drag on the orbit of planetesimals was investigated in detail by Adachi *et al.* (1976) by using a perturbation method. Using their formula, Tanaka and Ida (1997) studied the distribution of planetesimals around a protoplanet through semi-analytical calculation of the orbits of planetesimals using the Hill formalism (Tanaka and Ida 1996). They found that gravitational scattering by the protoplanet and the gas drag clean up the planetesimals in the feeding zone of the protoplanet. However, Tanaka and Ida (1997) did not include the effect of the mutual gravitational interaction of planetesimals and assumed equal-mass planetesimals. It is necessary to study the evolution of planetesimals in the solar nebula by using a more realistic model where the mutual gravitational interaction and the mass distribution of planetesimals are taken into account. Note that we include only the hydrodynamic drag of gas. Tidal interaction of protoplanets with the solar nebula is not included, the effect of which on orbital evolution and accretion is potentially important (e.g., Ward 1986, 1997; Tanaka and Ida 1999). The inclusion of this effect is a future subject.

In the present paper, we investigate the evolution of planetesimals in the solar nebula through N-body simulation. We take into account the gas drag on planetesimals by the nebula gas. We summarize the method of calculation in Section 2. The results of N-body simulations are presented in Section 3. The results are essentially the same as the previous gas-free simulations. Through runaway and oligarchic growth, a bimodal protoplanet–planetesimal system is formed, where similar-sized protoplanets grow keeping a certain orbital separation, while planetesimals remain small. Section 4 is devoted to summary and discussion.

2. METHOD OF CALCULATION

2.1. Model

We adopt the similar model to the minimum-mass solar nebula model (Hayashi 1981) for the solar nebula. The density of the gas ρ and the surface density of the solid material Σ of the solar nebula are given by

$$\rho = 2 \times 10^{-9} \left(\frac{a}{1\ \mathrm{AU}} \right)^{-11/4} (\mathrm{g\ cm^{-3}}), \tag{1}$$

$$\Sigma = 10 \left(\frac{a}{1\ \mathrm{AU}} \right)^{-3/2} (\mathrm{g\ cm^{-2}}), \tag{2}$$

where a is the distance from the sun and Σ is valid for the terrestrial planet region ($a < 2.7$ AU) where ice is not condensed. Our model is 50% more massive than the minimum-mass solar nebula model. The gas component of the solar nebula rotates with the velocity slightly slower than the Keplerian circular velocity $\mathbf{v}_\mathrm{c}$ due to the negative pressure gradient in the nebula gas (Adachi *et al.* 1976), which takes the form

$$\mathbf{v}_\mathrm{gas} = (1 - 2\eta)^{1/2} \mathbf{v}_\mathrm{c}, \tag{3}$$

where η is given as

$$\eta = 0.0019 \left(\frac{a}{1\ \mathrm{AU}} \right)^{1/2}. \tag{4}$$

We assume that there are no turbulent motions in the nebula.

The radius of planetesimals is given by

$$r = 2.3 \times 10^{7} \left(\frac{m}{10^{23}\ \mathrm{g}} \right)^{1/3} \left(\frac{\rho_\mathrm{p}}{2\ \mathrm{g\ cm^{-3}}} \right)^{-1/3} (\mathrm{cm}), \tag{5}$$

where m is the mass and ρ_p is the internal density of planetesimals. The drag force on a planetesimal per unit mass takes the form

$$\mathbf{f}_{gas} = -\frac{1}{2m} C_D \pi r^2 \rho |\mathbf{u}|\mathbf{u}, \tag{6}$$

where C_D is the drag coefficient and $\mathbf{u}$ is the relative velocity between the planetesimal and the gas. In our calculation, we adopt $C_D = 2$ and $\rho_p = 2\ \mathrm{g\ cm^{-3}}$, which are suitable for planetesimals. For simplicity, we assume that ρ_p is independent of mass. The magnitude of the drag force compared with the dominant solar gravitational force f_{sun} is estimated as

$$\frac{f_{gas}}{f_{sun}} \simeq 3 \times 10^{-8} \left(\frac{m}{10^{23}\ \mathrm{g}}\right)^{-1/3} \left(\frac{\rho}{2 \times 10^{-9}\ \mathrm{g\ cm^{-3}}}\right) \times \left(\frac{e}{0.01}\right)^2 \left(\frac{a}{1\ \mathrm{AU}}\right), \tag{7}$$

where we used $u \simeq (e^2 + i^2)^{1/2} v_c$ and $e = 2i$ and a, e, and i are the semimajor axis, the eccentricity, and the inclination of a plantesimal, respectively. This estimate shows that the magnitude of the drag force is large enough to be evaluated accurately in the double precision format (15 effective digits) for planetesimals considered in the present paper ($10^{23} \lesssim m \lesssim 10^{26}$ g).

Adachi *et al.* (1976) investigated the damping of the semimajor axis, the eccentricity, and the inclination of a planetesimal due to the gas drag force by using a perturbation method. We performed a numerical integration of a two-body system directly calculating the drag force by the Hermite scheme (see the next subsection) and found that the decay rates of the orbital elements obtained by the numerical integration are in good agreement with those of Adachi *et al.* (1976).

The eccentricities and inclinations of planetesimals are pumped up by the viscous stirring among themselves, while gas drag and collisions damp their eccentricities and inclinations. The equilibrium eccentricity and inclination of planetesimals with mass $m \gtrsim 10^{20}$ g are estimated by equating the time scales of the viscous stirring and the gas drag (Ohtsuki *et al.* 1993). For simplicity, we think of an equal-mass planetesimal system. The time scale of the gas drag is given by

$$T_{gas} \equiv \frac{u}{f_{gas}} \simeq 3.2 \times 10^2 (e^2 + i^2 + \eta^2)^{-1/2} \left(\frac{m}{10^{23}\ \mathrm{g}}\right)^{1/3} \times \left(\frac{\rho}{2 \times 10^{-9}\ \mathrm{g\ cm^{-3}}}\right)^{-1} \left(\frac{a}{1\ \mathrm{AU}}\right)^{1/2} \text{(years)}, \tag{8}$$

where a, e, and i are the semimajor axis, the eccentricity, and the inclination of a planetesimal and we used $u \simeq (e^2 + i^2 + \eta^2)^{1/2} v_c$.

The time scale of the viscous stirring is given by

$$T_{stir} \equiv \frac{v}{dv/dt} \simeq \frac{2}{n \pi r_g^2 v \ln \Lambda} \simeq 3 \times 10^{14} i (e^2 + i^2)^{3/2} \times \left(\frac{m}{10^{23}\ \mathrm{g}}\right)^{-1} \left(\frac{\Sigma}{10\ \mathrm{g\ cm^{-2}}}\right)^{-1} \left(\frac{a}{1\ \mathrm{AU}}\right)^{-1/2} \text{(years)}, \tag{9}$$

where v is the random velocity (deviation velocity from the local Keplerian circular velocity), n is the number density of planetesimals, r_g is the 90° deflection radius, and we used $r_g = 2Gm/v^2$, $n \simeq \Sigma/(m 2ia)$, and $v \simeq (e^2 + i^2)^{1/2} v_c$ (Ida 1990). The factor $\ln \Lambda$ comes from the effect of distant encounters and for planetesimal systems; it takes about 3 (Ida 1990, Stewart and Ida 2000). Equating these two time scales, we obtain the equilibrium eccentricity and inclination

$$e_{eq} = 2 i_{eq} \simeq 4.3 \times 10^{-3} \left(\frac{m}{10^{23}\ \mathrm{g}}\right)^{4/15} \left(\frac{\Sigma}{10\ \mathrm{g\ cm^{-2}}}\right)^{1/5} \times \left(\frac{\rho}{2 \times 10^{-9}\ \mathrm{g\ cm^{-3}}}\right)^{-1/5} \left(\frac{a}{1\ \mathrm{AU}}\right)^{1/5}, \tag{10}$$

where we assumed $e = 2i$ and $e^2 \gg \eta^2$. Figure 1 shows the time evolution of the RMS eccentricity and inclination for a system of 3000 equal-mass (10^{23} g) planetesimal ring with the ring width $\Delta a = 0.02$ AU around 1 AU. We used the boundary condition presented in Section 2.3. For this system, the RMS equilibrium

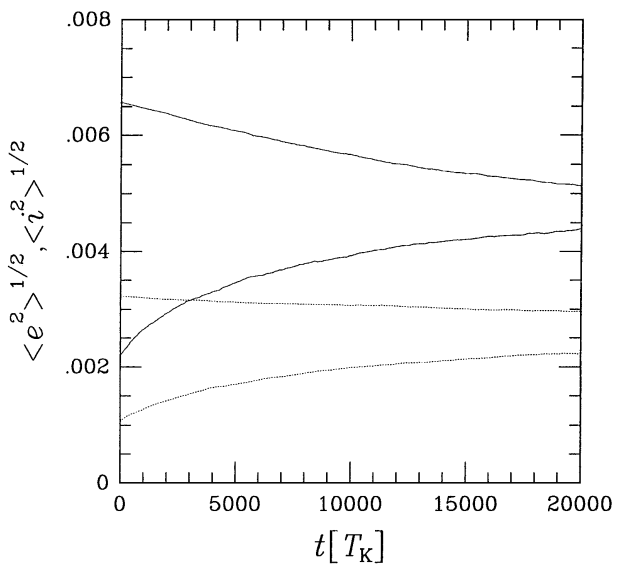

FIG. 1. Time evolution of the RMS eccentricity and inclination of a 3000 equal-mass (10^{23} g) planetesimal system. The initial values are 50% larger and smaller than the estimated equilibrium values. The solid curves represent the RMS eccentricity and the dotted curves the RMS inclination.

eccentricity and inclination are estimated as $e_{\rm eq} = 2i_{\rm eq} \simeq 0.0043$. The initial RMS eccentricity and inclination are 50% smaller and larger than the equilibrium values, respectively. The result of the N-body simulation agrees well with the above analytical estimate. The RMS eccentricity and inclination approach the equilibrium values from both higher and lower values. We also found that the distributions of the eccentricity and the inclination near the equilibrium values agree well with the Rayleigh distribution.

2.2. Initial Condition and Orbital Integration

Planetesimals are initially distributed in a ring around $a = 1$ AU with the width of $\Delta a = 0.02$–0.09 AU. The initial eccentricities and inclinations of planetesimals are given by the Rayleigh distribution (Ida and Makino 1992a).

The orbits of planetesimals are calculated by numerically integrating the equation of motion of planetesimals

$$\frac{d\mathbf{v}_i}{dt} = -GM_\odot \frac{\mathbf{x}_i}{|\mathbf{x}_i|^3} - \sum_{j\neq i}^{N} Gm_j \frac{\mathbf{x}_i - \mathbf{x}_j}{|\mathbf{x}_i - \mathbf{x}_j|^3} + \mathbf{f}_{\rm gas}, \qquad (11)$$

where $\mathbf{x}$ and $\mathbf{v}$ are the position and velocity of planetesimals respectively, G is the gravitational constant, and $M_\odot$ is the solar mass. The r.h.s. of Eq. (11) represents the solar gravity, the mutual gravitational interaction of planetesimals, and the gas drag force from left to right.

For the numerical integration, we use the modified Hermite scheme for planetary N-body simulation (Kokubo *et al.* 1998) with the hierarchical timestep (Makino 1991) with the maximum timestep of 2^{-5} (the Kepler period at 1 AU is 2π in the units we use).

The decay rate of the semimajor axis by the gas drag is given by (Adachi *et al.* 1976)

$$\frac{da}{dt} \simeq -1.3\times 10^{-2}\left(\frac{5}{8}e^2 + \frac{1}{2}i^2 + \eta^2\right)^{1/2}\left(e^2 + \frac{1}{8}i^2 + \eta\right) \times \left(\frac{m}{10^{23}\ \mathrm{g}}\right)^{-1/3}\left(\frac{\rho}{2\times 10^{-9}\ \mathrm{g\ cm^{-3}}}\right) \times \left(\frac{a}{1\ \mathrm{AU}}\right)^{1/2} (\mathrm{AU\ years^{-1}}). \qquad (12)$$

In our simulation, the smallest mass of a planetesimal is 10^{23} g. Assuming $e = 2i$, $e^2 \gg \eta^2$, and $\eta \gg e^2$ and using the typical value of our simulation $e \simeq 0.02$, we obtain $da/dt \sim -10^{-7}$ AU years^{-1} for the smallest planetesimal at 1 AU. On the other hand, the growth rate of the error in the semimajor axis due to the integration error of the Hermite scheme is smaller than 10^{-8} AU years^{-1} with the timestep of 2^{-5} (for details, see Kokubo *et al.* 1998). Thus, the accuracy of the scheme is high enough to follow the effect of the gas drag.

The most expensive parts of the Hermite scheme is the calculation of the mutual gravitational force and its time derivative, the cost of which increases in proportion to the square of the number of bodies. We calculate the force and its first time derivative by directly summing up interactions of all pairs on the special-purpose computer for N-body simulation, HARP-3/GRAPE-4 (Makino *et al.* 1993, 1997), whose peak speed is about 1 Tflops.

For simplicity, we assume that two planetesimals always accrete when they contact. In accretion, the position and the velocity of the center of mass are conserved. The lack of collisional fragmentation or rebound seems to make no significant change in the growth mode of protoplanets (Wetherill and Stewart 1993). The accretion of collisionally fragmented bodies may accelerate the accretion of protoplanets in the late stages. The inclusion of fragmentation is, however, beyond the present work and left for a future work.

2.3. Boundary Condition

The planetesimal system concerned here is a narrow ring with a width $\Delta a = 0.02$–0.09 AU. Without any boundary conditions, the surface density of planetesimals decreases considerably by the gas drag that damps the semimajor axis of a planetesimal as well as its eccentricity and inclination. It also decreases through the diffusion due to mutual gravitational interaction among planetesimals. This reduction of the surface density in the ring is unrealistic since in a realistic disk there is supply of bodies from the neighborhood. For example, in our simulation, as calculated in the previous subsection, the decay rate of the semimajor axis due to the gas drag is $da/dt \sim -10^{-7}$ AU years^{-1} for the smallest planetesimal at 1 AU. This means in the time scale of the present calculation 10^5 years, a substantial amount of planetesimals may fall toward the sun, which greatly reduces the surface density of the ring.

In the present work, we keep the surface density of the planetesimal ring constant by adopting the following "body-supply" boundary condition. The boundary condition is as follows: when the semimajor axis of a planetesimal becomes smaller/larger than the inner/outer edge of the ring, we take away the planetesimal and then create a new planetesimal with the same mass on the other edge. The eccentricity and the inclination of the new planetesimal are chosen so that they fit the local RMS values.

It is also possible to use the distance from the sun for the boundary index instead of the semimajor axis. However, if the distance from the sun is used, a body near the boundary may often go back and forth across the boundaries due to the epicyclic motion. This jumping of the boundaries may cause the unrealistic heating of the disk around the boundaries.

The semimajor-axis-based boundary condition affects the velocity distribution near the boundaries. This is because the semimajor-axis-based boundary condition removes the bodies that have a semimajor axis smaller/larger than the inner/outer boundary but have an eccentricity large enough to enter the ring at near apocenter/pericenter. As a result, the averaged tangential velocity becomes larger/smaller than the local Keplerian circular velocity near the inner/outer boundary. The averaged radial and vertical velocities are not affected by the boundary condition. An

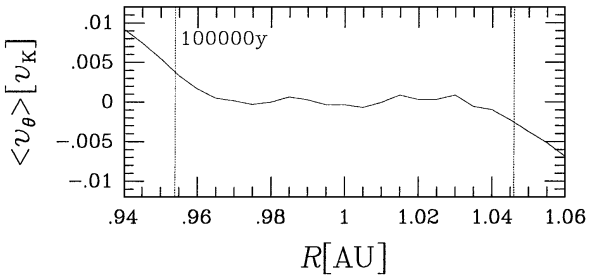

FIG. 2. The averaged tangential component of the random velocity of planetesimals $\langle v_\theta \rangle$ against the distance from the sun R for the wide range calculation at $t = 100{,}000$ years. The dotted lines show the boundaries of the semimajor axis based boundary condition.

example of the velocity anomaly is shown in Fig. 2 where the averaged tangential component of the random velocity of bodies is plotted against the distance from the sun (not the semimajor axis of bodies) for the wide range calculation at $t = 100{,}000$ years (see Section 3.2). The velocity anomaly exerts a positive/negative torque to the ring near the inner/outer boundary, which tends to shrink the ring. In our calculation, however, the decay of the semimajor axis of bodies due to gas drag and the diffusion of bodies surpass this effect.

The artificial torque tends to repel protoplanets near the boundaries. This repulsion is effective within the distance $\langle e^2 \rangle^{1/2} a$ from the boundary, where $\langle e^2 \rangle^{1/2}$ is the RMS eccentricity of bodies. In our wide range simulation, $\Delta a = 0.092$ AU and $\langle e^2 \rangle^{1/2} \simeq 0.02$, which means about 40% of the ring suffers from the artificial torque. On the other hand, adopting a finite ring omits the perturbation to protoplanets near the boundary from protoplanets outside the ring that exist in a realistic disk. This effect causes an artificial negative/positive torque near the inner/outer boundary. Fortunately, the artificial torques due to adopting the semimajor-axis-based boundary condition and the finite ring make up for each other. In other words, the artificial torque due to the boundary condition may play the same role as the torque from protoplanets outside the ring. The time scale of repulsion by the boundary is on the same order as the orbital repulsion between two protoplanets (G. R. Stewart 1999, private communication). It should be noted that we should make the ring as wide as possible to reduce the effects of the finite ring and the boundary condition as far as the computational power permits.

3. FROM PLANETESIMALS TO PROTOPLANETS

We present the results of N-body simulations in the runaway and the oligarchic growth stages and investigate the growth from planetesimals to protoplanets. In the runaway growth stage, the largest planetesimal grows in the runaway mode. In the oligarchic growth stage, protoplanets (runaway planetesimals) interact with each other and the similar-sized protoplanets are formed with a certain orbital separation. These results are essentially the same as those of N-body simulations with gas-free accretion and artificial enhancement of physical size (Kokubo and Ida 1996, 1998), although the accretion time scale is longer in the present calculation owing to adopting a realistic physical size.

In the following, we present the results of N-body simulations of planetary accretion from the intermediate stage. We start with planetesimals with mass $\sim 10^{23}$ g for convenience of calculation (the inclusion of smaller planetesimals, which might be important, is a future subject).

3.1. Runaway Growth

We present the result of the N-body simulation of the planetesimal system that initially consists of 3000 equal-mass ($m = 10^{23}$ g) bodies. The width of the planetesimal ring is $\Delta a = 0.021$ AU. The initial dispersion of the eccentricity and the inclination of bodies are given by the equilibrium values estimated by Eq. (10), and $\langle e^2 \rangle^{1/2} = 2\langle i^2 \rangle^{1/2}$ is assumed.

Figure 3 shows the snapshots of the system on the a–e plane for $t = 0$, 50,000, 100,000, 200,000 years. The filled circle indicates a protoplanet whose mass is larger than 100 times the initial mass. In 200,000 years, the number of bodies decreases to 1322. It is shown that a planetesimal grows rapidly. In Fig. 4, we plot the maximum mass and the mean mass of the system against time. The mass ratio between the maximum and the mean masses increases with time. At $t = 200{,}000$ years, the maximum mass reaches about 200 times the initial mass, while the mean mass becomes only about twice larger, which means that most mass remains in the smallest (initial) bodies as will be discussed below.

The evolution of the mass distribution is shown in Fig. 5. The cumulative number of bodies n_c against mass is plotted. The mass distribution relaxes to the distribution that is well approximated by a power-law distribution. The largest body at 200,000 years is shown by a dot that is separated from the continuous mass distribution. The mass range $10^{23} \le m \le 10^{24}$ g can be approximated by $dn_c/dm \propto n \propto m^\alpha$, where n is the number of bodies in a linear mass bin. The power indices calculated by using the least-square-fit method are $\alpha = -2.6$ for $t = 100{,}000$ years and $\alpha = -2.2$ for $t = 200{,}000$ years. These values are consistent with the results of the gas-free calculation by Kokubo and Ida (1996) and references therein. Since the total mass is $\int nm\,dm \propto m^{\alpha+2}$, $\alpha < -2$ means that small bodies contain most of the system mass.

The power index gradually decreases with time since there is no supply of small bodies in our simulation. The power index smaller than -2 is a characteristic of runaway growth under the perfect accretion; that is, most mass of the system exists in small bodies, while the largest body becomes more massive. We define the growth mode that produces a power-law mass distribution with the power index $\alpha < -2$ as runaway growth. Kokubo and Ida (1996) used the term runaway growth in the sense that the ratio between the maximum and mean masses of a system increases with time. This usage does not seem appropriate since this ratio always starts to increase when the distribution evolves from an equal-mass system. However, what Kokubo and Ida (1996) found is runaway growth still by the present definition because mean mass is always similar to the smallest mass and

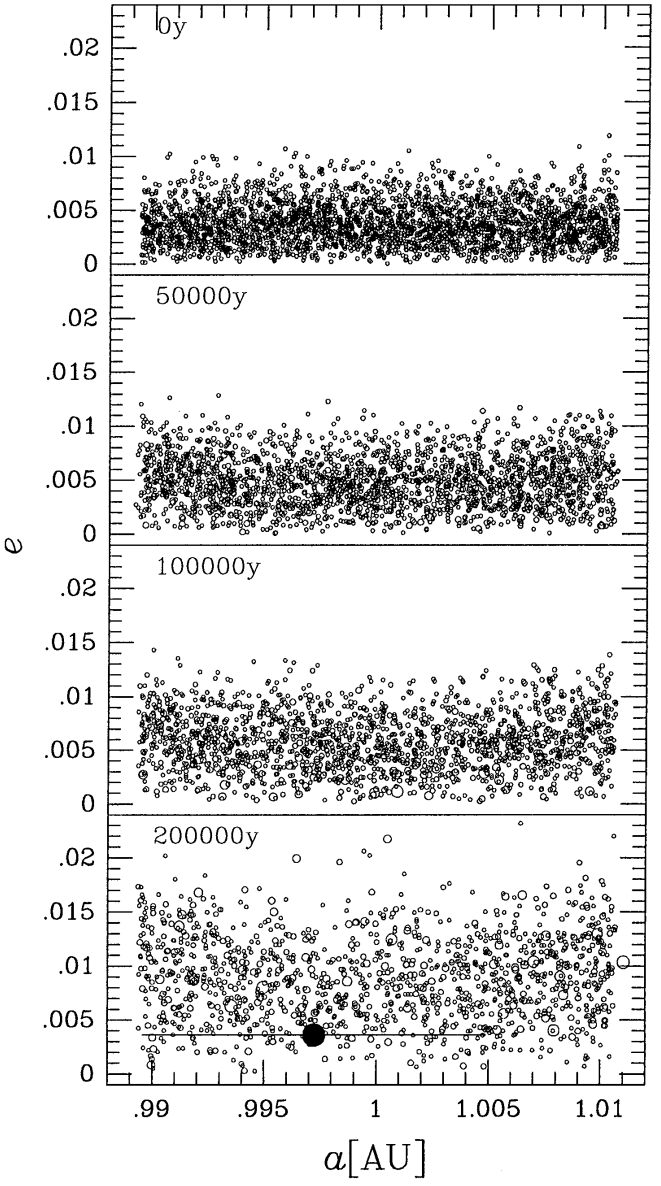

FIG. 3. Snapshots of a planetesimal system on the a–e plane. The circles represent planetesimals and their radii are proportional to the radii of planetesimals. The system initially consists of 3000 equal-mass (10^{23} g) planetesimals. The numbers of planetesimals are 2215 ($t = 50{,}000$ years), 1787 ($t = 100{,}000$ years), and 1322 ($t = 200{,}000$ years). In the $t = 200{,}000$ years panel, the filled circle represents a protoplanet (runaway body) and lines from the center of the protoplanet to both sides have the length of $5r_{\mathrm{H}}$.

the mass ratio keeps increasing when $\alpha < -2$. If $\alpha > -2$, mean mass should be similar to the maximum mass, so that the increase in the mass ratio would stall soon. Note that our definition of runaway growth does not necessarily mean that the growth time decreases with the mass of a body, but that the mass ratio of any two bodies increases with time as shown below.

The evolution of the distributions of the RMS eccentricity and the RMS inclination is plotted in Fig. 6. Let us focus on the mass range $10^{23} \le m \le 10^{24}$ g. The values for the mass range larger

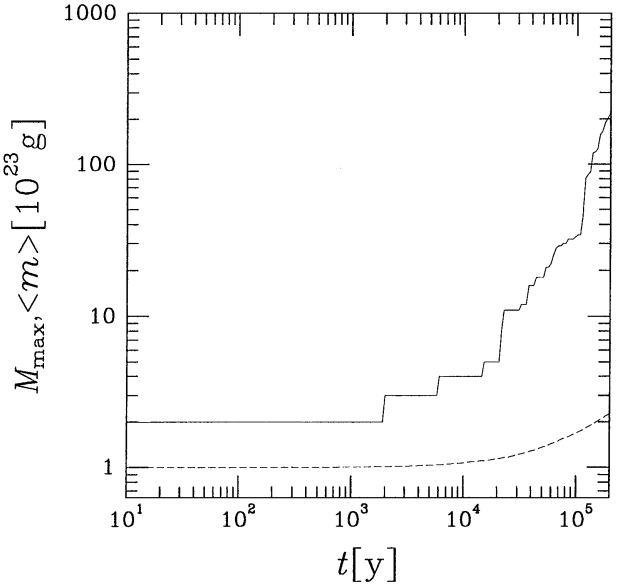

FIG. 4. Time evolution of the maximum mass (solid curve) and the mean mass (dashed curve) of the system.

than this range are not statistically valid since each mass bin often has only a few bodies. First, the distribution tends to relax to a decreasing function of mass through dynamical friction among (energy equipartition of) bodies ($t = 50{,}000$, $100{,}000$ years). Second, the distributions tend to flatten ($t = 200{,}000$ years). This is because as a runaway body grows, the system is mainly heated by the runaway body (Ida and Makino 1993). In this case, the eccentricity and inclination of planetesimals are scaled by the

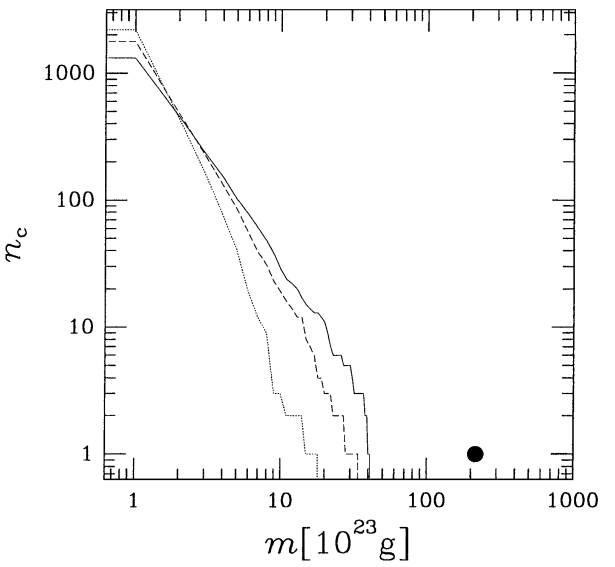

FIG. 5. The cumulative number of bodies is plotted against mass at $t = 50{,}000$ years (dotted curve), $100{,}000$ years (dashed curve), and $200{,}000$ years (solid curve). A runaway body at $t = 200{,}000$ years is shown by a dot.

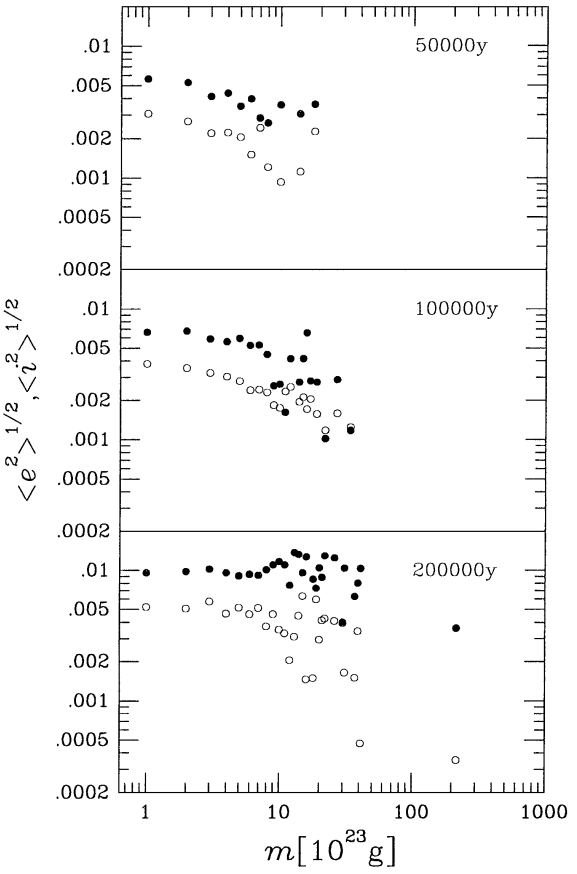

FIG. 6. Time evolution of the RMS eccentricities (filled circles) and inclinations (open circles) for each mass bin.

Hill radius of the runaway body and the scaled values are almost independent of the mass of planetesimals. In addition, as the gas drag is stronger for small bodies with high eccentricity and inclination, the eccentricity and the inclination of small bodies are more damped by the gas drag than those of large bodies, which also leads to the flat distributions of the eccentricity and the inclination. Note that the eccentricity and the inclination of the largest body are always kept small due to dynamical friction from smaller bodies. Small eccentricity and inclination facilitate runaway growth (Wetherill and Stewart 1989, Ohtsuki and Ida 1990, Kokubo and Ida 1996).

We have done four runs with different random numbers for the initial distribution of planetesimals. The detailed way in which a runaway body is formed differs in runs. However, the results are the same in the sense that a runaway body whose mass is about 200 times the initial mass is formed after 200,000 years.

The relation between the mass and the random velocity (eccentricity and inclination) distributions is derived by the simple analytical argument when the mass and the random velocity distributions are approximated by power-law distributions. Let us assume that the mass distribution is proportional to m^α and the random velocity distribution m^β. The stationary power-law state of the mass distribution is realized when the flux

$$nm\frac{dm}{dt} = \text{constant}, \tag{13}$$

which means not the number flux but the mass flux is independent of mass (for details, see Makino *et al.* (1998)). When the mass distribution is considered, the growth rate of a planetesimal is given by

$$\frac{dm}{dt} = \int_0^m \frac{n_s(m')}{\max(H, H')}\pi(r+r')^2\left(1+\frac{v_{\rm esc}^2}{v_{\rm rel}^2}\right)v_{\rm rel}m'\,dm', \tag{14}$$

where n_s is the surface number density, H is the scale height of the planetesimal system, and $v_{\rm esc}$ and $v_{\rm rel}$ are the surface escape velocity of the planetesimal and the relative velocity between planetesimals, respectively. When gravitational focusing is effective, the growth rate reduces to

$$\frac{dm}{dt} \propto m^{\alpha-2\beta+10/3}, \tag{15}$$

where we used $H \simeq 2ai$ and $v_{\rm rel}^2 \simeq v_{\rm ran}^2 + v_{\rm ran}'^2$, $v_{\rm ran}$ being the random velocity of planetesimals. From Eq. (13), for the stationary state, the relation between the mass and the velocity distributions is given by

$$\alpha = \beta - \tfrac{13}{6}. \tag{16}$$

This relation gives $\alpha = -\frac{8}{3}$ for $\beta \simeq -\frac{1}{2}$ (energy equipartition). In our calculation, $\beta \simeq -\frac{1}{2}$ at 100,000 years and the corresponding power index α agrees well with the above estimate. This argument assumed the supply of planetesimals at the low-mass end and the removal of planetesimals at the high-mass end. In our calculation, the former assumption does not hold, while a runaway body removes planetesimals at the high-mass end. Therefore, as the number of bodies decreases, the distribution becomes more gentle due to the lack of the supply of planetesimals at the low-mass end.

In the stationary state, the relative growth rate is given as $(1/m)\,dm/dt \propto m^{\alpha-2\beta+7/3}$. Substituting Eq. (16) into this relation yields

$$\frac{1}{m}\frac{dm}{dt} \propto m^{-\alpha-2}. \tag{17}$$

This shows that our definition of runaway growth $\alpha < -2$ is equivalent to the definition that the relative growth rate increases with mass; that is, the mass ratio of any two bodies increases with time. It should be noted that from the above argument the sufficient condition for runaway growth taking into account the mass distribution of planetesimals is $\beta < \frac{1}{6}$. This criterion corresponds to the limiting case of Ohtsuki *et al.* (1993) where gravitational focusing is dominant.

The runaway body keeps growing and then is isolated from the continuous power-law mass distribution. In this stage, the runaway body predominantly grows in its feeding zone as a sink of the mass flow from the continuous power-law mass distribution.

We confirmed that the growth mode of planetesimals is runaway growth (the power index of the mass distribution $\alpha < -2$) by the N-body simulation including the effect of gas drag. The effect of gas drag does not break the physical conditions for runaway growth: gravitational focusing is effective and dynamical friction is effective (the approximated power index of the velocity distribution $\beta < \frac{1}{6}$). In the present simulation, the effect of gas drag is damping the velocity dispersion and the semimajor axes of planetesimals. The evolution of the mass distribution is almost the same as that of gas-free accretion. The continuous distribution of the low mass part is well approximated by the power-law distribution $n \propto m^{\alpha}$, where $\alpha \simeq -2.5$. The eccentricity and inclination of a runaway body is always kept small due to dynamical friction from small planetesimals. Note that we adopted no radius enhancement. The time scale presented here is realistic.

3.2. Oligarchic Growth

Protoplanets are formed through runaway growth of planetesimals. We are now interested in the late runaway stage where protoplanets grow while interacting with each other. Kokubo and Ida (1998) investigated this stage and found oligarchic growth of protoplanets: similar-sized protoplanets grow, keeping their orbital separation larger than $5r_{\rm H}$, where $r_{\rm H}$ is the Hill radius of the protoplanets defined by

$$r_{\rm H} = \left(\frac{2M}{3M_{\odot}}\right)^{1/3} a, \tag{18}$$

where M is the mass of the protoplanet, while most planetesimals remain small. Among protoplanets, larger ones grow more slowly than smaller ones, while the relative growth rate of protoplanets is still larger than that of planetesimals. This growth mode is due to the slowdown of runaway growth (Ida and Makino 1993). While the protoplanets grow, orbital repulsion (Kokubo and Ida 1995) keeps their orbital separations wider than about $5r_{\rm H}$ of the protoplanets. For the orbital separation larger than $5r_{\rm H}$ and small eccentricities $e \lesssim h$, the Jacobi energy is negative and thus protoplanets cannot reach within their Hill radius (e.g., Hayashi *et al.* 1977). Kokubo and Ida (1998) studied only the gas-free case. Here we investigate the late runaway stage taking into account the effect of gas drag and using realistic-sized planetesimals.

We present the result of the N-body simulation of the planetesimal system that initially consists of 4000 bodies. The initial mass distribution is given by the power-law mass distribution with the power index $\alpha = -2.5$ with the mass range $2 \times 10^{23} \le m \le 4 \times 10^{24}$ g. The planetesimal ring is located at 1 AU with the width of $\Delta a = 0.092$ AU. The total mass in the ring is 1.3×10^{27} g, which is consistent with the surface density $\Sigma = 10$ g cm^{-2}. The initial dispersion of the eccentricity and the inclination of bodies are given by $\langle e^2 \rangle^{1/2} = 2\langle i^2 \rangle^{1/2} = 0.002$. We have performed two runs with different random numbers for the initial distribution of planetesimals. The results are essentially the same.

Figure 7 shows an example of the snapshots of the system on the a–e plane for $t = 100{,}000$, 200,000, 300,000, 400,000, 500,000 years. The corresponding mass distributions are plotted in Fig. 8. We plot the differential mass distribution (numbers of bodies in each mass bin) because it is easy to see individual protoplanets. In 500,000 years, the number of bodies decreases to 1257. The filled circles in Fig. 7 represent protoplanets whose masses are larger than 100 times the minimum mass of the system, 2×10^{25} g. The evolution of the system is summarized as follows:

$t = 100{,}000$ years. Four protoplanets are formed through runaway growth. The eccentricities and the inclinations of the protoplanets are kept small due to dynamical friction from small bodies. Their orbital separations are wider than about $5r_{\rm H}$. The minimum separation of about $5r_{\rm H}$ is kept by orbital repulsion between neighboring protoplanets.

$t = 200{,}000$ years. Seven protoplanets whose mass are on the same order have emerged. As they grow, their Hill radius increases and then it becomes difficult to keep the orbital separation by orbital repulsion since there is no room to expand it without shrinking other orbital separations. They start to interact with each other. Some protoplanets have relatively high eccentricity due to the recent scattering among protoplanets. These protoplanets with high eccentricity sometimes collide with other protoplanets.

$t = 300{,}000$ years. Two large protoplanets with mass $m \simeq 10^{26}$ g dominate the system. In between these two protoplanets, three small protoplanets with mass 3–4 $\times 10^{25}$ g have relatively large eccentricity.

$t = 400{,}000$ years. A large protoplanet is formed by the collision of two of the three small protoplanets. Now the system is dominated by three large protoplanets with mass 1–2 $\times 10^{26}$ g on nearly circular orbits. The orbital separation among these three protoplanets is 5–10$r_{\rm H}$, which is the result of orbital repulsion. The mass of the other protoplanets is 4–5 $\times 10^{25}$ g.

$t = 500{,}000$ years. The final state of the simulation is three large protoplanets with mass 1–2 $\times 10^{26}$ g on nearly circular noninclined orbits with an orbital separation of 5–10$r_{\rm H}$. Forty-one percent of the system mass is contained in the three large protoplanets.

Note that during $t = 300{,}000$–500,000 years, no large protoplanets cross the ring boundaries. In this stage, the eccentricities of the large protoplanets are kept small, because they are well separated and dynamical friction is still effective. The index α of the mass distribution approximated by power law for the mass range $2 \times 10^{23} \le m \le 10^{24}$ g calculated by the least-square-fit method decreases from -2.5 to -1.9 in 500,000 years since

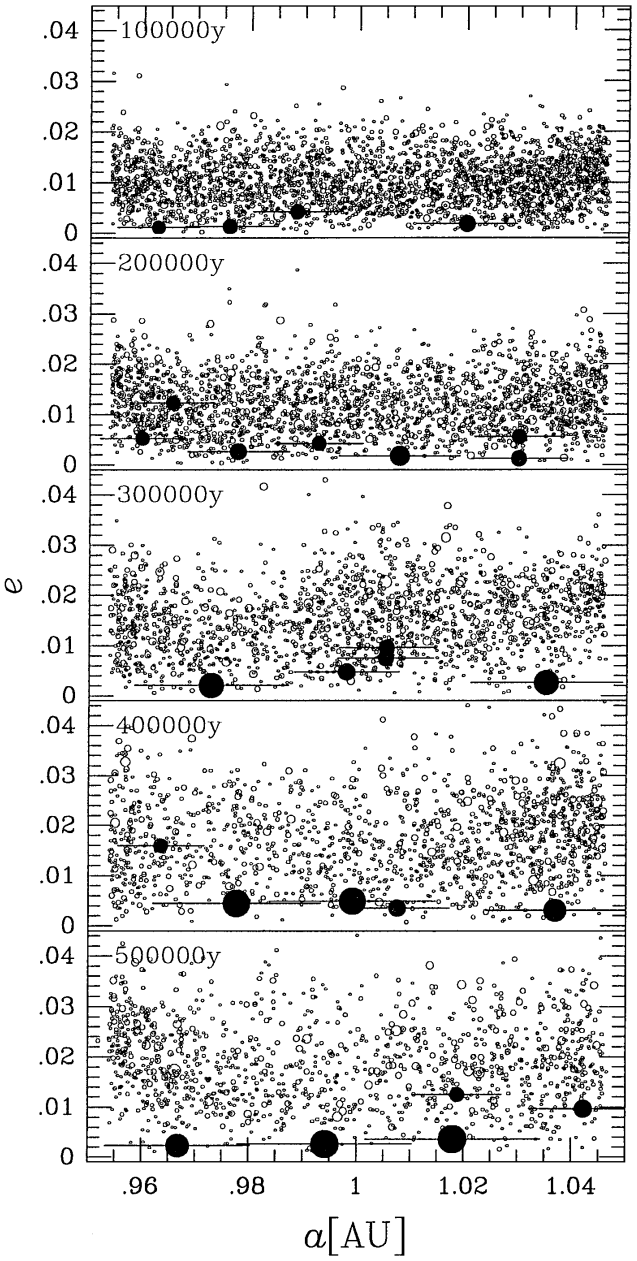

FIG. 7. Snapshots of a planetesimal system on the a–e plane. The circles represent planetesimals and their radii are proportional to the radii of planetesimals. The system initially consists of 4000 planetesimals whose total mass is 1.3×10^{27} g. The initial mass distribution is given by the power-law mass distribution with the power index $\alpha = -2.5$ with the mass range $2 \times 10^{23} \leq m \leq 4 \times 10^{24}$ g. The numbers of planetesimals are 2712 ($t = 100{,}000$ years), 2200 ($t = 200{,}000$ years), 1784 ($t = 300{,}000$ years), 1488 ($t = 400{,}000$ years), and 1257 ($t = 500{,}000$ years). The filled circles represent protoplanets with mass larger than 2×10^{25} g and lines from the center of the protoplanet to both sides have the length of $5r_{\mathrm{H}}$.

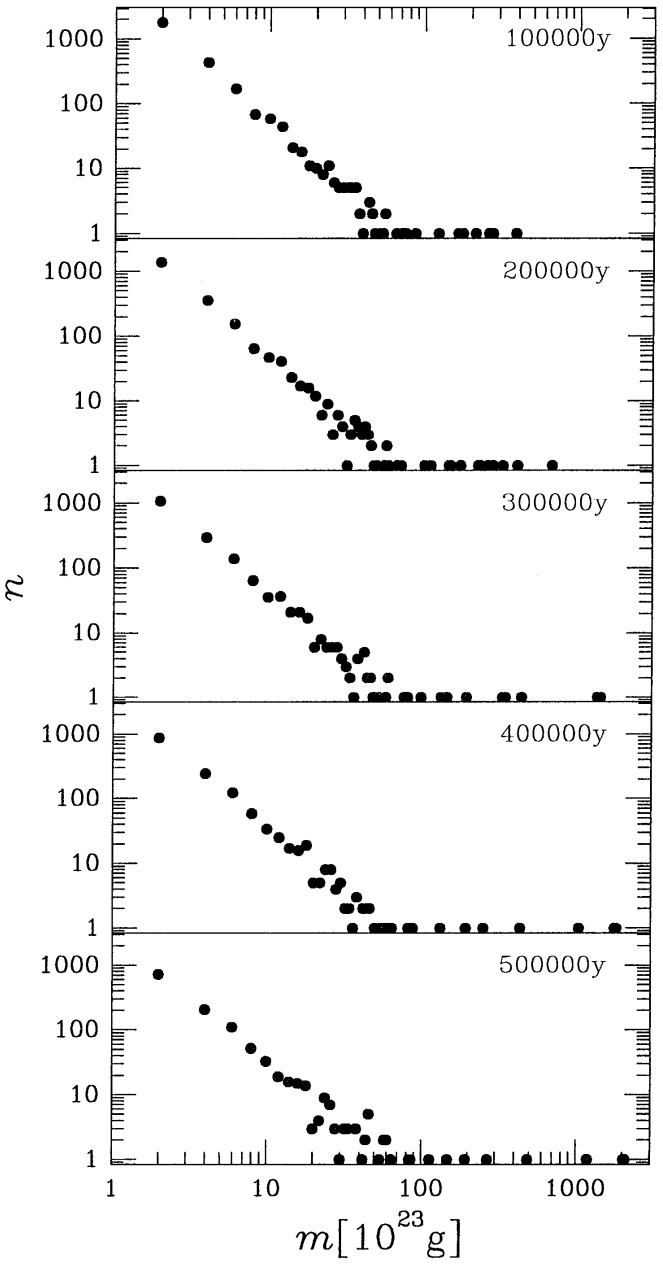

FIG. 8. The number of bodies in linear mass bins is plotted for $t = 100{,}000$, 200,000, 300,000, 400,000, and 500,000 years.

there is no supply of small bodies. Figure 9 shows the snapshot of another run at $t = 500{,}000$ years. Two large protoplanets with mass 1–3 $\times 10^{26}$ g with an orbital separation of about $10r_{\mathrm{H}}$ are formed. The two protoplanets contain 38% of the system mass.

In Fig. 10, we plot the maximum mass and the mean mass of the system against time. The time evolution of the RMS eccentricity and inclination of the system is plotted in Fig. 11. The values are scaled by the reduced Hill radius of the maximum body. The reduced Hill radius is given by $h = r_{\mathrm{H}}/a$. After about 20,000 years, the scaled RMS eccentricity and inclination are almost constant, $\langle e^2 \rangle^{1/2} \simeq 6h_{\max}$ and $\langle i^2 \rangle^{1/2} \simeq 3.5h_{\max}$. This result agrees with the estimation of Ida and Makino (1993). They estimated the equilibrium eccentricity and inclination of a planetesimal system perturbed dominantly by a protoplanet under gas drag. They found that when the mass of a protoplanet becomes

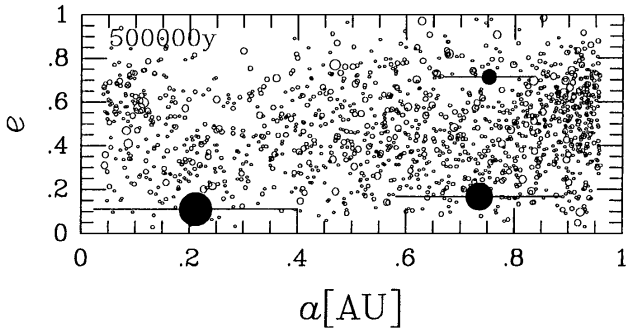

FIG. 9. A snapshot of a planetesimal system at $t = 500{,}000$ years on the a–e plane. The initial condition is the same as that described in the legend to Fig. 7. The number of planetesimals is 1385.

50–100 times larger than that of planetesimals with $m = 10^{23}$–10^{24} g, the velocity dispersion of planetesimals is dominantly determined by the protoplanet. In our calculation, the maximum mass that corresponds to the mass of the dominant perturbers exceeds 100 times the smallest mass of planetesimals around 20,000 years. In this stage, the growth of protoplanets slows down as the velocity dispersion of planetesimals in the vicinity of larger protoplanets is larger (Ida and Makino 1993).

The final outcome of the simulations is a few large protoplanets with mass 1–3×10^{26} g on nearly circular noninclined orbits with an orbital separation of 5–$10 r_{\mathrm{H}}$. This result is consistent with the oligarchic growth theory by Kokubo and Ida (1998). Similar-sized protoplanets are formed because among protoplanets, large ones tend to grow more slowly than small ones, since the large protoplanets increase the velocity dispersion of local planetesimals more effectively than small ones

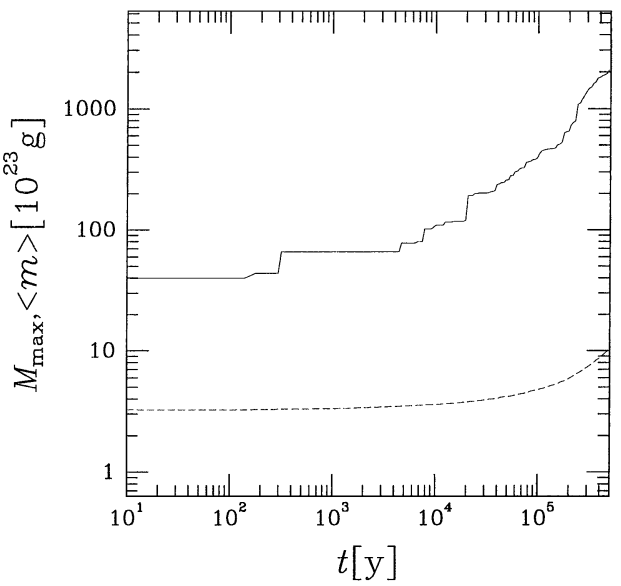

FIG. 10. Time evolution of the maximum mass (solid curve) and the mean mass (dashed curve) of the system.

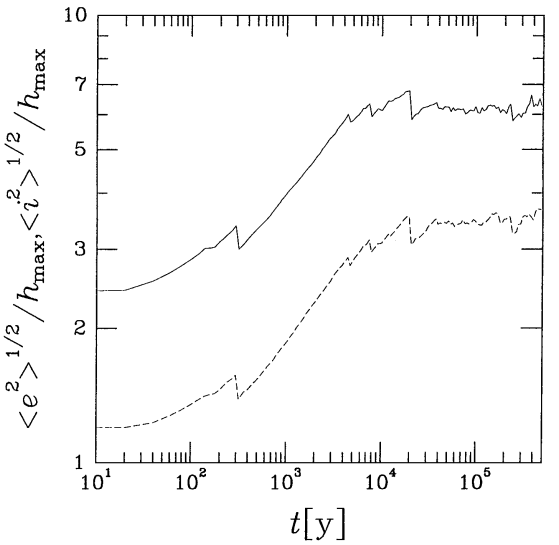

FIG. 11. Time evolution of the RMS eccentricities (solid curve) and inclinations (dashed curve) scaled by the reduced Hill radius of the maximum body $h_{\max}$.

and then the collision probability with those planetesimals decreases. The relative growth rate of protoplanets is, however, still larger than that of planetesimals. Thus, a bimodal protoplanet–planetesimal system is formed. Protoplanets grow while interacting with neighbor protoplanets. Through orbital repulsion (Kokubo and Ida 1995), they always expand the orbital separation larger than about $5 r_{\mathrm{H}}$. The collision time scale of protoplanets is usually larger than the time scale of orbital repulsion.

The mass of protoplanets growing in the oligarchic growth mode is estimated by

$$M = 2\pi a b p \Sigma, \tag{19}$$

where b is the orbital separation of protoplanets and p is the ratio of the mass of protoplanets to the total mass of a system, which gives

$$M \simeq 0.16 p^{3/2} \left(\frac{b}{10 r_{\mathrm{H}}}\right)^{3/2} \left(\frac{\Sigma}{10\ \mathrm{g\ cm^{-2}}}\right)^{3/2} \left(\frac{a}{1\ \mathrm{AU}}\right)^{3} M_{\oplus}\ (\mathrm{E}), \tag{20}$$

where $M_{\oplus}$ is the Earth mass and E stands for the terrestrial planet region ($a < 2.7$ AU). The growth time scale of the protoplanet is given by

$$T_{\mathrm{grow}} \equiv \frac{M}{dM/dt} \simeq 2 \times 10^{4} \left(\frac{e}{h_M}\right)^{2} \left(\frac{M}{10^{26}\ \mathrm{g}}\right)^{1/3} \times \left(\frac{\Sigma}{10\ \mathrm{g\ cm^{-2}}}\right)^{-1} \left(\frac{a}{1\ \mathrm{AU}}\right)^{1/2} \text{years (E)}, \tag{21}$$

where h_M is the reduced Hill radius of the protoplanet given by $h_M \simeq (M/3M_\odot)^{1/3}$ and $e = 2i$ is assumed (e.g., Kokubo and Ida 1995). For the result at $t = 500{,}000$ years, $p \simeq 0.5$. The above estimation gives $M \simeq 3 \times 10^{26}$ g for $p = 0.5$, which is consistent with the result of the N-body simulation. For $M = 10^{26}$ g at 1 AU, $T_{\rm grow} \simeq 7 \times 10^5$ years with $e = 6h_M$, which also agrees with the result of the N-body calculation.

The final mass of protoplanets gained through oligarchic growth is given by setting $p = 1$ in Eq. (20). For example, the final mass is $0.16M_\oplus$ at 1 AU. We discuss the final mass of protoplanets in the next section.

Our results are in good agreement with the results of the multizoned coagulation equation (Weidenschilling *et al.* 1997). In their larger simulation with $\Sigma = 8.4(a/{\rm AU})^{-3/2}$ g cm^{-2}, a few protoplanets with mass several times 10^{26} g are formed around 1 AU in 1,000,000 years, where 88% of the total mass is contained in the protoplanets. Their orbital separations are $\sim 10r_{\rm H}$. Their result agrees well with the above oligarchic growth theory. Substituting $\Sigma = 8.4$ g cm^{-2}, $b = 10r_{\rm H}$, and $p = 0.88$, Eq. (20) gives $M \simeq 6 \times 10^{26}$ g, whose growth time scale is given as $T_{\rm grow} \simeq 2 \times 10^6$ years from Eq. (21). The general agreement of the results of the N-body simulation and the multizoned coagulation equation suggests that the approach of solving the coagulation equation can follow planetary accretion accurately at least up to the stage of oligarchic growth. It is, however, necessary to perform a further detailed comparison with the N-body simulation to refine the coagulation equation model.

4. SUMMARY AND DISCUSSION

We have performed three-dimensional N-body simulations of planetary accretion from planetesimals to protoplanets. We extended N-body simulations to include the drag force due to the solar nebula gas. Differently from the previous N-body simulations (Kokubo and Ida 1996, 1998) that adopted the radius enhancement of planetesimals to save computational time, realistic-sized planetesimals with a standard material density 2 g cm^{-3} are used. We also adopted the boundary condition for a planetesimal ring that prevents mass decrease in the ring due to radial migration of planetesimals.

In agreement with the previous N-body simulations of gas-free planetary accretion (Kokubo and Ida 1996, 1998), we found that a bimodal protoplanet–planetesimal system is formed through runaway and oligarchic growth. The evolution of planetesimals is summarized as follows. In the intermediate stage, larger planetesimals grow faster than smaller ones, resulting in the runaway growth of the largest planetesimal. This is because in a planetesimal system, the larger a planetesimal becomes, the larger its growth rate becomes by gravitational focusing and dynamical friction from smaller planetesimals. In this stage, the mass distribution relaxes into the continuous power-law mass distribution with $dn_{\rm c}/dm \propto m^\alpha$ with $\alpha \simeq -2.5$. Then the largest body at the top of the continuous mass distribution becomes isolated from the continuous mass distribution. The distribution of velocity dispersion (eccentricity and inclination) first relaxes to a decreasing function of mass owing to dynamical friction and then becomes almost flat for low-mass bodies because runaway bodies dominate the heating of small bodies. Protoplanets (runaway planetesimals) are formed through runaway growth. As protoplanets grow, they start to interact with one another. By orbital repulsion (Kokubo and Ida 1995), they try to repel neighbors to a distance wider than 5 Hill radius ($5r_{\rm H}$). However, when there is no space to expand the separation without intruding into other orbital separations, some protoplanets are thinned out by collisions with other protoplanets. Then the growth mode shifts to oligarchic growth where larger protoplanets tend to grow more slowly than smaller ones, while the mass ratios between protoplanetes and planetesimals increase. As a result, in the late stage, similar-sized protoplanets grow oligarchically, while most planetesimals remain small. The mass distribution is divided into a small number of large protoplanets and a large number of small planetesimals. The protoplanets grow, keeping their orbital separations wider than about $5r_{\rm H}$ of the protoplanets by orbital repulsion as far as dynamical friction from small planetesimals is effective. The result of the N-body simulation shows that in 500,000 years, protoplanets with masses $\sim 10^{26}$ g are formed at 1 AU. This result agrees well with the analytical estimate based on the oligarchic growth theory. Our results also generally agree with the results of the multizoned coagulation equation (Weidenschilling *et al.* 1997).

Based on the oligarchic growth theory, we can make a rough estimate of the final mass and growth time of protoplanets as Eq. (20) and Eq. (21). We extrapolate this estimation with $p = 1$ to the jovian planet region. The mass and growth time of protoplanets are given by

$$M \simeq 5\left(\frac{b}{10r_{\rm H}}\right)^{3/2}\left(\frac{\Sigma}{4\ {\rm g\ cm^{-2}}}\right)^{3/2}\left(\frac{a}{5\ {\rm AU}}\right)^{3} M_\oplus\ ({\rm J}), \quad (22)$$

$$T_{\rm grow} \simeq 9 \times 10^4\left(\frac{e}{h_M}\right)^{2}\left(\frac{M}{10^{26}\ {\rm g}}\right)^{1/3}\left(\frac{\Sigma}{4\ {\rm g\ cm^{-2}}}\right)^{-1} \times \left(\frac{a}{5\ {\rm AU}}\right)^{1/2}\ {\rm years\ (J)}, \quad (23)$$

where J stands for the jovian planet region ($a > 2.7$ AU). From Eqs. (20), (21), (22), and (23), we obtain the final mass, orbital separation, and growth time of protoplanets at 1, 5, 10, 20, and 30 AU which correspond to the positions of Earth, Jupiter, Saturn, Uranus, and Neptune (Table I), where we assumed $b = 10r_{\rm H}$,

$$\Sigma = \begin{cases} 10\left(\dfrac{a}{1\ {\rm AU}}\right)^{-3/2}\ {\rm g\ cm^{-2}\ (E)} \\ 4\left(\dfrac{a}{5\ {\rm AU}}\right)^{-3/2}\ {\rm g\ cm^{-2}\ (J)} \end{cases}, \quad (24)$$

TABLE I
The Final Mass, Orbital Separation, and Growth Time of Protoplanets

a (AU)	Σ (g cm^{-2})	M ($M_\oplus$)	b (AU)	$T_{\rm grow}$ (10^6 years)
1	10	0.16	0.07	0.7
5	4	5	1	40
10	1.4	9	3	300
20	0.5	14	6	2000
30	0.3	20	10	7000

$$e \simeq \begin{cases} 6\left(\dfrac{m}{10^{23}\ \mathrm{g}}\right)^{1/18}\left(\dfrac{a}{1\ \mathrm{AU}}\right)^{7/24} h_M\ (\mathrm{E}) \\ 11\left(\dfrac{m}{10^{24}\ \mathrm{g}}\right)^{1/18}\left(\dfrac{a}{5\ \mathrm{AU}}\right)^{7/24} h_M\ (\mathrm{J}) \end{cases}, \qquad (25)$$

and $e = 2i$. We used a surface density 50% more massive than that in the minimum-mass nebula model (Hayashi 1981). The discontinuity of the surface density between E and J is due to the condensation of ice. The eccentricity of planetesimals is calculated by equating the stirring by protoplanets and the damping by gas drag (Ida and Makino 1993).

In the terrestrial planet region, the estimated mass and orbital separation of protoplanets are smaller than those of the present planets. This result suggests that the final stage of the terrestrial planet formation is collisions of protoplanets while sweeping out residual planetesimals. A new problem here is whether it is possible to make planets like the present terrestrial planets from protoplanets within a lifetime of the solar system.

The stability of equal-mass protoplanet systems has been investigated by N-body simulations (Chambers *et al.* 1996, Yoshinaga *et al.* 1999). Yoshinaga *et al.* (1999) found that in a protoplanet system with an orbital separation of $8r_{\rm H}$ and eccentricities of $2h_M$, orbital crossing of protoplanets takes place in 10^5 years. Chambers and Wetherill (1998) performed a long-term N-body simulation of the accretion of protoplanets with an orbital separation of $7r_{\rm H}$. They found that with Jupiter and Saturn of the present mass and orbital elements, planets like the present terrestrial planets are formed within the age of the solar system through accretion of protoplanets where the eccentricities of the protoplanets are raised due to the secular perturbation by the jovian planets. The eccentricities of the planets, however, is significantly larger than those of the present terrestrial planets.

In the jovian planet region, the protoplanets formed through oligarchic growth may be consistent with the formation of the present planets. The estimated final mass of protoplanets at 5 AU is $5M_\oplus$. The growth time of the Jupiter core is as long as the observationally estimated life time of the solar nebula ($\sim 10^7$ years) (Strom *et al.* 1993, Zuckerman *et al.* 1995). It is not clear that the protoplanet with $5M_\oplus$ can onset the gas accretion onto the protoplanet and capture the massive gas envelop from the solar nebula in the lifetime of the solar nebula at present. The growth time of the Saturn core is substantially longer than the estimated lifetime of the solar nebula, which may lead to capturing less gas envelope from the disappearing solar nebula. This may be consistent with the fact that the amount of the gas envelope of Saturn is smaller than that of Jupiter, though their core masses are estimated as almost the same. As for Uranus and Neptune, which consist mainly of solid material, the estimated final mass of protoplanets and the orbital separation are almost consistent with their present values. The growth time is as long as the age of the Solar System. As discussed by Kokubo and Ida (1998), the orbital separation of jovian planets ($\sim 10r_{\rm H}$) may be explained along with the orbital repulsion theory. Note that the orbital separation realized by orbital repulsion is proportional to the distance from the Sun since the orbital separation is proportional to the Hill radius.

We should, however, be careful when we apply oligarchic growth to the jovian planet region. Oligarchic growth is obtained from the narrow ring simulation in which the semi-major axis is much larger than the width of the ring and the perturbation from the outside of the ring is omitted. It is uncertain that oligarchic growth takes place in the wide jovian planet region in the same way as the local area simulation. Further work on this issue is required.

The migration of protoplanets due to tidal interaction with the solar nebula proposed by Ward (1986, 1997) is a potentially important effect that is not included in the local area calculation. In fact, Tanaka and Ida (1999) showed that the inward migration of protoplanets accelerates the growth of protoplanets since planetesimals with low random velocity are supplied as protoplanets migrate. The migration of protoplanets is, however, still uncertain at present.

Our model does not take into account fragmentation of bodies. In the final stage, however, it must play an important role. Particularly, the accretion of small fragmented bodies whose velocity dispersion is small due to gas drag can accelerate the growth of protoplanets (Wetherill and Stewart 1993). The effect of inelastic collisions that do not result in accretion is not included either. However, in the mass region we simulated, gas drag is dominant over inelastic collisions in damping velocity dispersion.

We found that a small number of protoplanets with their orbital separation wider than $5r_{\rm H}$ are formed from a swarm of planetesimals in the solar nebula in the late accretion stage through runaway and oligarchic growth. For the final stage of planetary formation, we should consider this protoplanet–planetesimal system as the initial condition. It should be noted that a substantial amount of small planetesimals may fill the space between protoplanets. To explore the final accretion stage, we need long-term N-body simulations. It is also necessary to take into account the effects of fragmentation and inelastic rebound of planetesimals. The present N-body simulation treated a narrow planetesimal ring where global effects such as distant perturbation by giant planets are not taken into account. In order to generalize further

the oligarchic growth theory, we need to perform more global N-body simulations.

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